D1. Describing data: summaries and plots by variable type

Always report n and the number of missing values. Plot the data before choosing a summary.

Variable type	Summarise with	Plot with	R, course data
One variable			
Binary or nominal e.g. smoker, district	Counts and percentages. State the denominator: everyone, or only those with a known value.	Bar chart of counts or percentages	`count(hearts, district)` `geom_bar()`
Ordinal e.g. satisfaction 1 to 5	Counts and percentages, kept in category order. Median and IQR can be added.	Bar chart in category order	`count(hearts, satisfaction)`
Continuous, roughly symmetric e.g. SBP	Mean and standard deviation (SD)	Histogram; boxplot	`mean(sbp_0); sd(sbp_0)` `geom_histogram(binwidth = 5)`

Continuous, skewed
e.g. CRP
	Median and interquartile range (IQR, 25th to 75th percentile). Mean and SD are pulled by extreme values.	Histogram, often on a log scale; boxplot	`median(crp, na.rm = TRUE)`
			`quantile(crp, c(0.25, 0.75), na.rm = TRUE)`
			`scale_x_log10()`

Count e.g. clinic visits	Mean and variance; a rate (events per person-time) when follow-up time differs	Bar chart of each count value	`sum(clinic_visits) / sum(followup_months)`
Two variables			
Continuous by group e.g. SBP by district	The summary above, given separately for each group, with n per group	Boxplots with the raw points	`group_by(district) |> summarise(...)` `geom_boxplot() + geom_jitter()`
Two categorical e.g. smoker by high BP	Contingency table with row percentages (denominator = each row total)	Bar chart of proportions	`tab <- table(hearts$smoker, hearts$high_bp_0)` `prop.table(tab, margin = 1)`
Two continuous e.g. age and SBP	Correlation, if the pattern is roughly linear or monotonic (see D4)	Scatterplot	`geom_point()`